



SEMESTER 1/20252026

**SETP4583- SURFACTANT AND DETERGENT
TECHNOLOGY**

INDIVIDUAL REPORT

SECTION EB01

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1. Introduction

At the beginning of this course, I assumed that detergents were relatively simple consumer products, mainly focused on removing dirt and grease. My initial expectation was that the course would primarily involve basic chemical concepts and straightforward laboratory activities. I chose to take this course to fulfil my academic requirements and to gain some general knowledge related to chemistry-based consumer products. At that time, my goal was to understand how detergents work at a basic level and to complete the course successfully.

However, as the course progressed, my perception of detergent science changed significantly. I began to realise that detergent formulation is a complex process that involves a balance between chemistry, functionality, safety, sustainability, and consumer perception. Through lectures, laboratory sessions, and the group project, I learned that even small formulation decisions can greatly affect product performance and appearance. This course challenged my initial assumptions and encouraged me to think more critically about everyday products that are often overlooked.

2. Reflection on Key Learnings

One of the most important concepts I learned in this course was the role of surfactants and how different types contribute to cleaning performance. I gained a clear understanding of anionic, amphoteric, and non-ionic surfactants, as well as how their interactions influence properties such as foaming, mildness, and stability. Before this course, I was unaware that detergents rely on carefully designed surfactant systems rather than a single active ingredient.

Laboratory sessions played a crucial role in reinforcing theoretical knowledge. By formulating a dishwashing liquid prototype, I learned how ingredient ratios and mixing order affect the final product. During the experiment, I observed changes in colour, clarity, and viscosity as different surfactants were added. One key learning moment occurred when our detergent formulation turned milky instead of transparent. Initially, I believed this indicated an error in the preparation process. However, through further discussion and research, I learned that turbidity can result from the formation of mixed surfactant structures rather than incomplete dissolution. This experience helped me understand that appearance does not necessarily reflect product effectiveness.

The group project further enhanced my understanding by requiring us to justify ingredient selection and evaluate product performance. Working collaboratively allowed me to learn from my teammates' perspectives, especially when discussing formulation rationale and testing outcomes. I also developed a better appreciation for how laboratory experimentation complements theoretical concepts, turning abstract ideas into observable results.

Overall, this course helped me move beyond memorisation of concepts and toward analytical thinking. I learned to interpret experimental observations using scientific reasoning and to accept that unexpected results can still be valid and meaningful.

3. Application and Real-World Relevance

The knowledge gained from this course has clear real-world relevance, particularly in the detergent and consumer goods industry. I now understand that commercial detergent products are designed not only for cleaning performance but also for safety, environmental impact, and consumer acceptance. For example, many consumers associate transparent detergents with higher quality, even though non-transparent formulations can perform equally well. This insight highlights how formulation decisions are influenced by marketing and consumer perception as much as by chemistry.

The course also increased my awareness of sustainability considerations in product development. Learning about biodegradable surfactants and toxicity assessment made me realise the importance of selecting ingredients that minimise environmental harm. In real-world applications, manufacturers must consider wastewater treatment compatibility, regulatory requirements, and long-term ecological effects. This perspective helped me connect classroom learning to broader societal and environmental responsibilities.

Additionally, the laboratory experience reflects real industry practices, where formulations often require multiple trials and adjustments. The process of testing, observing, and refining a product mirrors how commercial detergents are developed and optimised. As a consumer, I now view household cleaning products more critically, paying attention to ingredient lists, safety labels, and sustainability claims.

4. Challenges and Growth

One of the main challenges I faced in this course was dealing with unexpected experimental outcomes. During the detergent formulation lab, the product's milky appearance initially caused confusion and self-doubt, as I assumed that a correct detergent should be transparent. This challenge forced me to confront my limited understanding and to seek explanations rather than immediately labelling the result as a failure.

Through further learning and reflection, I realised that this challenge represented a turning point in my growth. I learned to rely on scientific reasoning instead of assumptions and to evaluate results based on stability and performance rather than appearance alone. This experience improved my problem-solving skills and increased my confidence in handling experimental uncertainty.

Another challenge was balancing teamwork and individual responsibility during the group project. Coordinating ideas, dividing tasks, and ensuring consistency across sections required effective communication and time management. Over time, I became more proactive in contributing ideas and clarifying my assigned responsibilities, which helped me grow as a team member.

5. Personal and Professional Development

This course has influenced my personal and professional mindset by encouraging a more analytical and reflective approach to learning. I have developed a stronger foundation in detergent science and a greater appreciation for the complexity of product formulation. The emphasis on safety and sustainability has also shaped my perspective on responsible product development.

Professionally, the course has enhanced my ability to analyse real-world problems using scientific principles. Even if I do not pursue a career directly related to detergents, the skills gained—such as critical thinking, experimentation, and reflection—are transferable to many fields. In the future, I would like to explore sustainability-focused product design further and continue developing my understanding of how science can be applied to improve everyday products.

Conclusion

In conclusion, this course has significantly expanded my understanding of detergent science and challenged my initial assumptions. Through lectures, laboratory work, and group collaboration, I developed both technical knowledge and reflective skills. The experience taught me that learning often occurs through challenges and unexpected outcomes, making the course a valuable and meaningful part of my academic journey.

Appendix



Class Photo



Group Photo

